

Python Lists - Complete Notes

1. What is a List?

Definition: A list is an ordered, mutable collection that can store multiple values of different data types.

Syntax:

```
nums = [10, 20, 30]
fruits = ["apple", "mango"]
mixed = [1, "Python", True]
```

Features: Ordered Mutable (changeable) Allows duplicates Supports indexing and slicing

2. Indexing

Access elements using positive and negative indexes.

```
nums[0], nums[-1]
```

3. Iteration

Using **for** and **while** loops to visit every element.

4. Adding Elements

`append(x)`: add one item at end.

`insert(i,x)`: insert at position.

`extend(lst)`: add multiple items.

5. Updating Elements

```
nums[1] = 99
```

6. Removing Elements

`remove(value)`, `pop(index)`, `clear()`

7. Useful Methods

`count(x)`: frequency

`index(x)`: first position of value

8. Sorting

`sort()` ascending

`sort(reverse=True)` descending

`reverse()` reverses current order

9. Copy

`copy()` creates independent copy.

Using `'=`' creates another reference to the same list.

10. Slicing

nums[start:end:step]

Examples: nums[:3], nums[2:], nums[::-1], nums[::-2]

11. Nested Lists

Example: students=[["Vipin",90],["Rahul",85]]

Access: students[0][1]

12. Nested Loops

```
for row in matrix:
```

```
    for value in row:
```

```
        print(value)
```

13. List Comprehension

Basic: [i for i in range(5)]

Filtering: [i for i in range(10) if i%2==0]

Conditional: ['Even' if i%2==0 else 'Odd' for i in range(5)]

14. Searching

in / not in

index(value): position

count(value): occurrences

Manual search using loops.

15. DSA Tips

Practice: Find largest/smallest without max()/min() Sum elements Count even/odd Reverse a list
Second largest Remove duplicates Frequency counting

16. Methods Summary

append(x) - add one

extend(lst) - add many

insert(i,x) - insert

remove(x) - remove by value

pop(i) - remove by index

clear() - remove all

index(x) - first index

count(x) - occurrences

sort() - ascending

reverse() - reverse order

copy() - independent copy